Ek Tasveer — one trusted picture of every child's foundational learning

A NIPUN Bharat decision-support layer built on UDISE+, PARAKH, ASER and state assessments

India @ 71/100 Challenge · Theme 2: Foundational Learning Outcomes · Round 1 proposal — draft v0.3 (4 Oct 2026)

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1. Problem identification

The scorecard. India scores 71/100 on the SDG India Index 2023-24 and 61 on SDG 4 (Quality Education); the adjusted net enrolment rate in elementary education is 96.5 % [1]. NIPUN Bharat (launched 5 July 2021) commits to "universal FLN in primary schools by 2026-27": every child reads with meaning at **≥ 60 words a minute** and handles numbers to 9,999 by the end of Grade 3 [2]. **2026-27 is this school year.**

The lived reality. In rural **government** schools, **23.4 %** of Std III children can read a Std II-level text and **27.6 %** can do at least subtraction (ASER 2024) [3]. Both rose sharply from 2022 (16.3 % and 20.2 %), so the system *can* move — but three in four Grade 3 children in government schools still cannot read a Grade 2 text. In private schools the reading figure is 35.5 % [3].

The measurable gap. *Share of Grade 3 children in government schools who meet the Grade 3 reading and numeracy goals*: mission target 100 % in 2026-27 [2]; independently measured ≈ 23 % (reading) and ≈ 28 % (subtraction) [3].

The underserved group. Children in Grades 1–3 of rural government schools — 68.9 % of India's 14.71 lakh schools are government schools [4] — and, within them, those the data system sees least well: children in **1,04,125 single-teacher schools (33.8 lakh enrolled)** [4], Scheduled Tribe children — the lowest-scoring social group in PARAKH 2024 Grade 3 (maths 57 % vs 61 % for others) [7] — Scheduled Caste children, and children whose home language is not the language of instruction.

The implementation gap — five breaks. Data is not missing: UDISE+ has held student-wise records ("more than 60 fields for each student") since 2022-23 [4]; states run census-style FLN tests; Vidya Samiksha Kendras (VSKs) run dashboards. Yet the chain *recorded → believed → acted on → rechecked* breaks at five points:

Break	Today	Consequence
Commensurability	PARAKH 2024 reports Grade 3 children answering ~64 % of language items correctly [7]; ASER 2024 reports 23.4 % of government-school Std III children reading a Std II text [3]. Both are right — they measure different things — but no system links them	Teacher, block and state see different pictures
Trust	Census-style tests are often "proctored and graded by teachers in the same school"; Madhya Pradesh's Pratibha Parv (~6–7 million students) was shown to "mislead inferences about learning levels" [5]	Officials cannot tell progress from reporting pressure

Break	Today	Consequence
Granularity	Trustworthy surveys (ASER, PARAKH) are samples	Silent about a block, a school, a child
Actionability	Outputs are averages and "% NIPUN"	Nobody is told *which children, which skill, which activity, this week*
Closed loop	Actions are not logged or rechecked	Nobody learns what worked

2. Evidence and context

Evidence	Finding	Source
ASER 2024 (rural, household, one-on-one oral test)	Std III govt: reading Std II text 20.9 % (2018) → 16.3 % (2022) → 23.4 % (2024); subtraction 20.2 % (2022) → 27.6 % (2024)	[3]
Foundational Learning Study 2022 (NCERT; ~86,000 Grade 3 students, 10,000 schools, 20 languages)	Numeracy: 37 % limited, 42 % sufficient, 10 % superior skills	[6]
PARAKH Rashtriya Sarvekshan 2024 (21,15,022 children, 74,229 schools, 781 districts; Grade 3: 5,99,026 students)	Grade 3 average score, language / maths: rural 64 % / 60 %; **ST children lowest at 61 % / 57 %** vs 66 % / 61 % for others; weakest maths competencies (shapes, money) at 50 %	[7]
Official-data audit, Madhya Pradesh (re-test in 283 schools)	Official census scores inflated by **+38.9 pp (maths)** and **+33.8 pp (Hindi)** over independent re-test levels of 25.1 % and 37.9 %; in Andhra Pradesh, paper tests inflated by 16–20 pp, while tablet-based tests (≈ US$0.50 per test) mostly removed the distortion	[5]
Diagnostic feedback to teachers, Andhra Pradesh (RCT)	Reports on their pupils' results alone: no detectable learning gain	[13]
School-governance reform, Madhya Pradesh (RCT, 1,774 schools)	High compliance with ratings and improvement plans; learning effects null (maths +0.031 SD, Hindi +0.004 SD)	[14]
Teaching at the Right Level (six RCTs, seven states)	Haryana +0.15 SD (language); Uttar Pradesh learning camps +0.6–0.7 SD; scaled to "over 60 million students"	[8], [15]
Phone + SMS targeted instruction, Botswana (RCT)	+0.121 SD numeracy; SMS alone +0.024 SD (not significant)	[16]
Global Education Evidence Advisory Panel (2023)	"Targeting teaching instruction by learning level, not grade" is a *Great Buy*	[17]

Design lesson. Reports and ratings produced compliance, not learning [13], [14]; unverified official numbers were off by over 30 points [5]; diagnosis that becomes a

grouped, level-appropriate action works [8], [15], [16]. So the layer must make numbers trustworthy and end every diagnosis in an action.

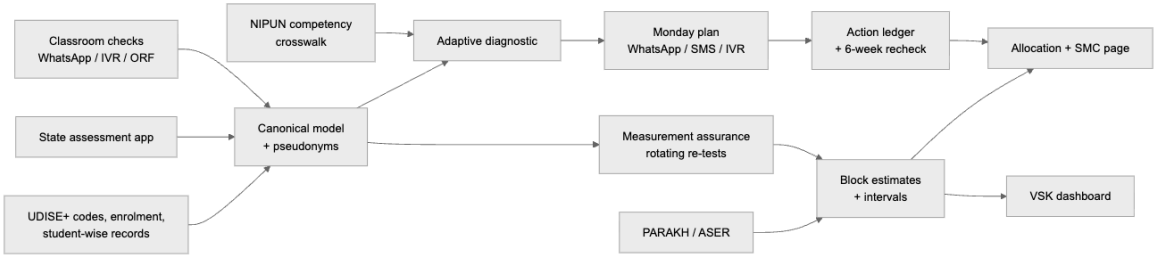
3. Existing ecosystem — why the gap persists

System (deployed)	What it does	What it leaves undone
UDISE+ [4]	School census + student-wise records since 2022-23	No learning levels
VSKs (Gujarat: 54,000 schools, 1.15 crore students, per officials [9])	Command-centre dashboards: attendance, assessments	Monitoring, not a classroom action; no causal evaluation found
State FLN assessments + verification — Haryana *Saksham Ghoshna*: a block is "Saksham" if > 80 % of students score > 50 %, checked by a third-party test in all 119 blocks [10]; Gujarat *Gunotsav*: 3,000 senior officials assess "25% schools from all the blocks" [19]; UP: NIPUN app with external verification [20]	Census data + verification	Verification grades schools or blocks; it is not used to *calibrate* the data, so it must be huge
AI oral reading — Wadhwani AI Vaachan Samiksha, ~30,000 Gujarat schools (developer figures) [11]	Tamper-resistant fluency scores	No independent accuracy check found; no remediation plan
TaRL (Pratham) [8]; chatbots such as SwiftChat in 14 states (vendor figures) [12]	Assess → group → activities; delivery channels to teachers	Needs a light, recurring diagnostic and follow-up at scale

Why the gap persists. Data pipes scaled; verified impact did not. Teacher-graded data is fragile, so states verify by brute force, and verification samples are not used as measurement. Dashboards monitor; few systems turn a diagnosis into next week's activity.

4. Proposed solution — "one picture, two loops"

Ek Tasveer is **not a new app**. It is a thin decision layer that plugs into what states run — UDISE+ for master data, the state assessment app for capture, the VSK for dashboards, DIKSHA for content, WhatsApp/SMS/IVR for delivery — and adds the five missing links through 12 components with typed, open contracts (JSON Schema), so a state can adopt one component at a time.



Loop 1 — the classroom, weekly. A teacher runs a few-minute oral check per child on the NIPUN-aligned ladder (letter → word → Grade 1 text → Grade 2 text; numbers → subtraction → division). An adaptive engine stops as soon as it is ≥ 90 % sure of the child's level and returns a **Monday plan**: TaRL groups, one activity per group from the state's own content, and the competency to recheck in four weeks — as one WhatsApp message, ≤ 3 SMS, or an IVR call in the local language. Names stay on the teacher's phone; the system holds only pseudonyms.

Loop 2 — the system, termly ("measurement assurance"). Exam boards already *moderate* school-assessed marks against independent evidence — e.g. NSW's NESA and Queensland's QCAA, which reviews "a selected sample" from every school [21]. We adapt that idea, plus survey back-checks and small-area estimation, to FLN census data, using the sample to *calibrate*, not to grade. Each term a small, rotating sample of schools — chosen by statistical uncertainty and decision value, always with a random component, so every school has a known inclusion probability — is re-tested by the cluster coordinator during the regular academic-support visit, who then coaches that group. A model-assisted estimator calibrates every school's reported results; small-area estimation with UDISE+ covariates then gives **block numbers with intervals that add up to the PARAKH/ASER anchor**. No school-level "trust label" is published: the aim is a true picture, not blame.

Decisions it feeds. CRC weekly visit plans; block/district allocation of remedial camps and training, ranked by expected children helped and weighted for single-teacher schools and SC/ST share; a one-page pictorial summary for each School Management Committee; interval-backed evidence for district Samagra Shiksha plans.

Governance and standards. Keys: UDISE+ school code, LGD codes, a salted pseudonym of the student ID; competency codes from the NIPUN Lakshya Soochi. Role- and purpose-bound access: child-level data only on the teacher's device; cells with < 10 children suppressed; school-level moderation data only for the moderation team. The DPDP Act 2023 requires verifiable parental consent for children's data and bars "tracking or behavioural monitoring of children" unless rules exempt a class of processing (s. 9) [18]; the design holds either way, with pseudonymised, purpose-bound server records and consent captured at enrolment where required. NIPUN asks for "IT based solutions … field level child wise monitoring" built on DIKSHA, UDISE+ and NDEAR standards [2] — this is that layer.

Working demo (real runs; synthetic children — no real child data used). A simulated district (6 blocks, ~600 government primary schools) in which ~30 % of schools inflate teacher-reported results; code, seeds and commands in the repository (`eval/run_eval.py`, `components/*/evaluate.py`).

Result (first runs, 4 Oct 2026)	Value
Adaptive check capped at 8 of 16 items: exact level accuracy (5 levels, both subjects, seeds 101–105)	0.551 — baseline for the auto-research loop
Monday plan for a class of 60, Hindi WhatsApp	659 bytes; SMS version 3 segments; IVR script
Block error of "% meeting goal", teacher-reported vs moderated (10 % of schools re-tested; 5 seeds)	4.09 ± 0.32 pp → 2.34 ± 1.09 pp
Detecting inflating schools (AUC) from public signals only	0.77
Governance policy tests (`components/governance/evaluate.py`)	36 / 36 pass

Honest limits. In a small district (120 schools, 12 re-tested; one seed) moderation *increased* block error (5.4 → 7.7 pp) and intervals under-covered, so the minimum effective sample and calibrated intervals are our next targets; allocation gains are not yet shown. The 5-seed figures used the simulator's default calibration (25 % / 30 %); the ASER-calibrated re-run (23.4 % / 27.6 %) is pending.

5. Feasibility and implementation roadmap

Phase	Scope	What happens	Exit criterion
0 (3 months)	1 district, 2 blocks, clusters staggered	Loop 1 in Grades 1–3; Loop 2 at a 10 % rotating sample; staggered start gives a causal comparison	Teachers use the plan weekly; moderated vs reported gap measured
1 (1 year)	1 state via its VSK	Connectors to the state assessment app and UDISE+; SMC pages; AWP&B evidence	Block estimates with intervals in the state's planning cycle
2 (2–3 years)	Multi-state	Open contracts and components offered as digital public goods	Adopted component-by-component

People and time (design estimates). Teacher: one short oral check per child per term and one message a week. CRC: at a 10 % rotating sample, a block of 100 schools needs 10 school re-tests per term — about 12 children each, inside visits the CRC already makes. Block/district: the existing monthly review uses the moderated numbers. **Costs:** no new devices; SMS/IVR as fallback; tablet-based re-testing was costed at about US$0.50 per test in the Andhra Pradesh study [5]. **Risks → mitigations:** teacher workload → short adaptive checks; language → per-language validation of items and AI scoring; connectivity → offline-first, SMS/IVR; gaming → rotation, random component, no published labels; privacy → pseudonyms, suppression, purpose limits.

6. Expected impact and scalability

Beneficiaries. Grade 1–3 children in government schools of the pilot district, then the state (counted from UDISE+ grade-wise enrolment). **Outcome indicators.** Share of Grade 3 children meeting the reading and numeracy goals at block level, with intervals, every term; teacher minutes per child; share of actions rechecked within six weeks; gap between reported and moderated figures. **Expected effect and scale.** TaRL-style grouping has strong RCT support [8]; the staggered pilot measures our effect rather than assuming it. No new hardware or app: the layer runs on the state's VSK, uses UDISE+ identifiers and speaks WhatsApp/SMS/IVR.

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